

Airborne Lead and Birth Outcomes: Evidence from Leaded Fuel in Auto Racing

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Abstract

The National Association for Stock Car Auto Racing (NASCAR) and Automobile Racing Club of America (ARCA) races continued to use leaded gasoline until 2007, despite the removal of leaded gasoline from on-road vehicles long ago. Exploiting this natural experiment, I study how airborne lead exposure affected birth outcomes using administrative birth records linked to race events at Texas Motor Speedway (TMS) in Fort Worth, Texas. Before the transition to unleaded fuel, mothers living within 25 km of TMS experienced lower infant birthweight, shorter gestation, and a higher probability of low birthweight associated with race exposure, while the corresponding birthweight and gestational-age estimates are near zero after the transition. To account for how lead accumulates in the body, I developed a pregnancy-specific physiologically based pharmacokinetic (PBPK) model that simulates maternal and fetal daily blood lead trajectories. I find that moving from the 25th to the 75th percentile of simulated blood lead in the second trimester is associated with an approximately 73-gram reduction in birthweight. Suggestive evidence points to gestational hypertension as a mechanism through which lead shortens gestation. These findings suggest that even a relatively low level of lead can have a meaningful effect on maternal and infant health.

JEL classifications: Q53, I18, J13, Q58

Keywords: Lead exposure, birthweight, prenatal health, NASCAR, leaded gasoline, PBPK model

1 Introduction

Lead (Pb) is one of the most well-established environmental neurotoxins and has been associated with numerous adverse outcomes including cognitive impairment, increased criminal behavior, and higher mortality (Talayero et al., 2023; Lanphear et al., 2019; Deryugina et al., 2019; Knittel et al., 2016; Stieb et al., 2012; Currie and Schmieder, 2008). Growing evidence on the harmful effects of lead prompted many countries to adopt policies aimed at reducing exposure beginning in the 1970s. One of the most consequential interventions was the phaseout of leaded gasoline from passenger vehicles which produced substantial public health gains. Despite this, lead is still present today in sources such as fuel used in piston-engine aircraft, paint in older housing, and drinking water. Notably, the National Association for Stock Car Auto Racing (NASCAR) and the Automobile Racing Club of America (ARCA), two of the largest racing organizations in the United States, continued to use leaded racing fuel until late 2006 to early 2007 before switching to unleaded gasoline. I exploit this transition as a quasi-experimental setting to estimate the effect of lead exposure on birth outcomes among mothers living near Texas Motor Speedway (TMS) in Fort Worth, Texas between 2003 and 2010. Before the transition, races generated concentrated bursts of airborne lead emissions surrounding the racetrack. After the transition, racing continued with unleaded fuel which provides a clean comparison to isolate the effect of lead exposure from other race-related activities.

I identify the effect of prenatal lead exposure entirely from within-TMS variation as the traditional cross-track comparison approach may produce unreliable estimates in this setting. Furthermore, differences across racetracks and the sociodemographic characteristics of surrounding communities make it difficult to construct a credible comparison group. Hence, the within-TMS design exploits four sources of variation in exposure among mothers living near the same racetrack. First, mothers living closer to TMS are likely to receive greater lead exposure as lead concentrations decline with distance from the track. Second, wind direction on race days is plausibly exogenous to maternal characteristics and residential sorting so mothers residing downwind of the racetrack are more exposed than mothers outside the lead plume. Third, the gestational timing of each race depends on the mother's conception date. Mothers who live the same distance from TMS but conceive just a few weeks apart may experience different patterns of race exposure across trimesters. Finally, emissions vary across races with caution flags, accidents, and race length which generates variation in the amount of lead released that is plausibly unrelated to maternal health.

To estimate the effect of lead exposure on birth outcomes, I construct a trimester-level exposure measure and estimate split-sample regressions comparing births before and after

the transition from leaded to unleaded racing fuel. This approach follows a difference-in-differences logic since identification rests on the assumption that factors correlated with race-day intensity such as traffic, noise, and pollution operate similarly across the leaded and unleaded periods. A negative and statistically significant relationship between lead exposure and birth outcomes in the leaded period and a null result in the unleaded period would be consistent with lead as the mechanism driving the effect. While the effect of lead can be estimated using a simple reduced-form equation, this approach does not account for how lead accumulates within a body, particularly during pregnancy. Lead can accumulate in maternal blood, tissues, and bone and be remobilized from bone back to blood in later trimesters. To account for these dynamics, I developed a pregnancy-specific Physiologically-Based Pharmacokinetic (PBPK) model that translates each mother's discrete race-day exposure pulses into daily blood lead trajectories over a 280-day pregnancy. The model compartmentalizes lead in the maternal blood, tissues, and bones and returns trimester-specific blood lead means, area under the curve (AUC), and peak burdens while capturing the remobilization of lead stored in maternal bones during her later trimesters.

Before the transition, each additional race during the second trimester lowers birthweight by 31 grams, or 0.9 percent of mean birthweight, and shortens gestation by about one day, with smaller reductions for races in the first and third trimesters. After the transition, when the same races continued without lead, the birthweight and gestational-age estimates are small and statistically insignificant. The pre-transition estimates are identified mainly from births in 2005-2006, after the track added a second major race weekend that shifted race exposure across pregnancies. Birthweight also rises with distance from the track before the transition, by 1.25 grams per kilometer, while this gradient disappears afterward. Mothers who were downwind on race days in the first trimester delivered infants about 31.5 grams lighter than mothers exposed to the same number of races who were never downwind, although the post-transition wind estimates do not uniformly conform to the placebo prediction. The PBPK model yields a consistent dose-response. Moving from the 25th to the 75th percentile of simulated blood lead in the second trimester lowers birthweight by about 73 grams. Most of the birthweight reduction operates through shorter gestation, and suggestive evidence points to gestational hypertension as a channel.

A large body of work documents that exposure to ambient pollution reduces birthweight, increases the risk of preterm birth, and worsens a wide range of other infant health outcomes ([Pedersen et al., 2013](#); [Currie, 2013](#); [Stieb et al., 2012](#); [Currie and Neidell, 2005](#)). A related literature also found that lead reduces educational attainment, lower adult earnings, and generate persistent socioeconomic disadvantages over the life cycle ([Talayero](#)

et al., 2023; Jenkins et al., 2022; Boyle et al., 2021; Aizer et al., 2018; Liu et al., 2014). Yet, less is known about the effects of prenatal exposure to localized sources of airborne lead and how these effects vary with the timing of exposure during pregnancy despite the fact that lead readily crosses the placenta and can affect fetal development in utero (Santana et al., 2023; Yüksel et al., 2022; Zhu et al., 2010). This paper contributes to these literatures in three ways. First, I provide causal evidence on the effects of a localized source of airborne lead on maternal and infant health using within-track variation in exposure. The design compares mothers living near the same racetrack and exploits variation in race timing, emissions, distance, and wind direction rather than relying on comparisons across communities surrounding different racetracks. Second, I incorporate a pregnancy-specific PBPK model into the empirical design to account for the accumulation and remobilization of lead over the course of pregnancy. The model maps discrete race-day exposures into daily blood lead trajectories and allows exposure in one trimester to affect blood lead levels in subsequent trimesters. Third, I measure exposure at the maternal census-tract level and incorporate race-day wind direction to distinguish mothers residing downwind of the racetrack from those outside the lead plume rather than assigning exposure based solely on proximity.

The remainder of the paper proceeds as follows. Section 2 provides background on leaded gasoline and NASCAR's and ARCA's switch to unleaded gasoline in 2007. It also introduces the four Texas racetracks used in the analysis, with Texas Motor Speedway (TMS) as the primary track. Section 3 describes the birth records, race data, and weather data. Section 4 describes how I construct race-level lead exposure measures, along with the PBPK model used to trace lead exposure over the course of pregnancy. Section 5 describes the empirical strategy, which uses variation in the timing of exposure, distance from TMS, and wind direction. Section 6 presents the main results and examines heterogeneity by distance and wind direction, the PBPK estimates, and maternal hypertension as a potential mechanism. Section 7 discusses the magnitude of the estimates and concludes. Finally, the Appendix discusses robustness and falsification tests and provides a monitor-based exposure measure as an alternative specification.

NASCAR & ARCA racetracks and maternal residential locations in Texas
Live births within 25 km of each track (2003-2010); dashed ring = 25 km



Figure 1. Red diamonds indicate the four Texas racetracks that hosted NASCAR or ARCA races relevant to the analysis: Texas Motor Speedway in Fort Worth, Texas World Speedway in College Station, Thunder Hill Raceway in Kyle, and Meyer Speedway in Houston. Blue circles indicate maternal residential locations, measured using census tract centroids, in the birth sample. The racetracks in College Station and Houston are legacy racetracks while the racetrack in Kyle hosted regional NASCAR races during the sample period for which the leaded-fuel status cannot be determined.

2 Background and Institutional Setting

2.1 Leaded Gasoline in NASCAR & ARCA Races

Leaded gasoline remained common in motorsports long after it disappeared from commercial passenger vehicles. Lead served as an octane booster, which allows engines to operate at very high compression ratios without engine knock. Prior to the transition, NASCAR and ARCA races used fuel containing approximately 4 to 5.2 grams of tetraethyllead per gallon ([Hollingsworth et al., 2025](#)). The organizations operate hundreds of races each year through both national and regional racing series. NASCAR's three national series are the Cup Series, Xfinity Series (currently called O'Reilly Auto Parts Series), and Camping World Truck Series (currently called Craftsman Truck Series) while ARCA's main national series is the Menards Series. NASCAR officially acquired ARCA in late 2018 and has since operated the ARCA Menards Series as part of NASCAR's regional division. NASCAR also operated regional series including the K&N Pro Series

West (KNPSW). These series follow a strict guideline on the type of vehicles and fuel used in each race. NASCAR and ARCA required all cars in a given series to use the same standardized fuel supplied by its fuel sponsor, Sunoco ([Hollingsworth and Rudik, 2021](#)). This standardization implies that variation in lead exposure arises from the timing and location of races rather than from differences in fuel composition across drivers or teams.

The Clean Air Act of 1970 established the federal framework for regulating air pollution in the United States. Beginning in the 1970s, the EPA introduced regulations to gradually reduce lead in gasoline, in part because of growing concerns about the health effects of lead exposure. By 1996, leaded gasoline was completely phased out for on-road vehicles in the United States. Racing vehicles, however, were exempt from these requirements and continued to use leaded gasoline. NASCAR and ARCA began transitioning to unleaded gasoline in 2006 and permanently switched their national racing series to unleaded fuel in 2007. NASCAR's Xfinity and Camping World Truck Series began using unleaded fuel on September 23, 2006 while the Cup Series continued using leaded gasoline through the end of the 2006 season and permanently switched to unleaded fuel on February 25, 2007. ARCA also switched to unleaded fuel beginning in 2007.¹

Given this staggered transition, I divide the sample into three windows: pre-transition, transitional, and post-transition. The pre-transition window includes births conceived before December 17, 2005 so that the third trimester ended before the Xfinity and Camping World Truck Series began using unleaded fuel on September 23, 2006. The post-transition window includes births conceived on or after March 1, 2007, ensuring that the entire pregnancy occurred after the Cup Series permanently switched to unleaded fuel on February 25, 2007.² The transitional window includes births conceived between December 17, 2005 and February 28, 2007, and is excluded from the cross-sectional sample to prevent gestational overlap between the pre- and post-transition periods.

2.2 Study Area

Texas Motor Speedway (TMS) is located in Fort Worth, Texas and is the primary venue in this study. The 1.5-mile quad-oval racetrack is situated on roughly 2.3 square miles and hosted its first major race in 1997 and has since become a major NASCAR active national venue. During the sample period (2003–2010), TMS typically hosted six NASCAR races

¹Throughout the sample, ARCA had only one race at TMS in 2010, and hence, the entire analysis rests on NASCAR races which I will mainly refer to throughout this paper.

²I set the post-transition cutoff at March 1, 2007 rather than February 26, 2007 as this provides additional buffer to guard against measurement error in conception date which is calculated from the imputed date of birth for the 2008–2010 births.

per year across the Cup, Xfinity, and Truck Series, with race weekends occurring in April, June, and November. Table 13 provides a full list of NASCAR and ARCA race schedule at TMS and Thunder Hill Raceway between 2003 and 2010.

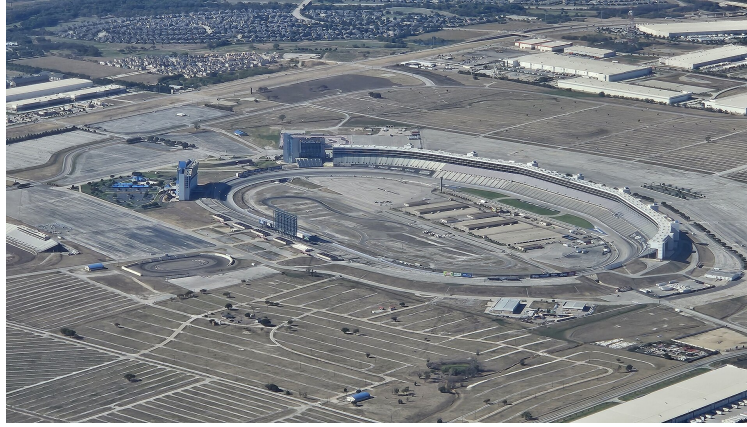


Figure 2. Note: Texas Motor Speedway in 2023 via Wikipedia

The second NASCAR venue, Thunder Hill Raceway, is located in Kyle, Texas, and hosted five K&N Pro Series West (KNPSW) races between 2005 and 2009. Because the KNPSW was a regional series, the timing of its transition to unleaded gasoline cannot be determined. Hence, I exclude Thunder Hill Raceway as a secondary leaded venue from the analysis. Texas World Speedway, located in College Station, and Meyer Speedway, located in Houston, are former NASCAR tracks whose last races occurred decades before the sample period (1993 and 1971, respectively). Table 1 summarizes the characteristics of the four Texas racetracks.

Table 1. NASCAR and ARCA Racetracks in Texas

Track Name	City	Lap Length	Period Active	Role
Texas Motor Speedway	Fort Worth	1.5 mi	1996–present	Primary
Thunder Hill Raceway	Kyle	0.375 mi	1998–2009	Excluded
Texas World Speedway	College Station	2.0 mi	1969–1993	Legacy track
Meyer Speedway	Houston	0.5 mi	1969–1971	Legacy track

Notes: Thunder Hill Raceway is excluded because the leaded-fuel status of its five regional NASCAR races between 2005 and 2009 cannot be determined.

3 Data

The analysis combines three primary data sources, which are linked at the individual birth level.

Birth records. I use confidential Texas birth records from 2003 to 2010, obtained from the Texas Department of State Health Services (DSHS) Vital Statistics. These records contain information on exact birth timing and location, birth outcomes, maternal characteristics, pregnancy complications, and the census tract of maternal residence. The primary outcome is birthweight in grams. Secondary outcomes include gestational age in weeks, low birthweight status ($< 2,500$ grams), and birthweight conditional on gestational age. In addition, the 2008-2010 birth records do not report the exact date of birth. For these years, I impute the birth date using the mother's reported last menstrual period and the LMP-based gestational length. Given that the data are provided for different years, I harmonize the variables across years and also restrict the sample to singleton births with an obstetric estimate of gestational age between 20 and 44 completed weeks.

Race data. Race data come from Racing Reference which contains every NASCAR and ARCA event held at the four Texas racetracks during the sample period: Texas Motor Speedway in Fort Worth, Meyer Speedway in Houston, Texas World Speedway in College Station, and Thunder Hill Raceway in Kyle. These data include the exact race date, track location, racing series, total laps, lap length, miles traveled, number of drivers, total race duration in seconds, and other race characteristics. The data distinguish across NASCAR's major racing series including the Cup Series, Xfinity Series, and Truck Series, which is important because different series transitioned away from leaded fuel at different times. For example, NASCAR Xfinity and NASCAR Craftsman Truck Series (CWTS) switched permanently to unleaded fuel on September 23, 2006 while NASCAR Cup Series switched to unleaded fuel on February 25, 2007. The sample also includes one ARCA race at TMS in 2010, which used unleaded fuel.

Weather data. Weather data come from the National Oceanic and Atmospheric Administration's (NOAA) NCEP-NCAR Reanalysis 1 model. Wind direction, wind speed, precipitation, air temperature, and relative humidity on each race day come from weather stations within 275 km of each track. Each station has an exact latitude and longitude that could be mapped to the birth and race data. For each birth $\times$ race day, I match to the nearest weather station and compute the bearing from the racetrack to the maternal census tract centroid. A mother is classified as "downwind" on race day j if the angle between the track-to-residence bearing and the observed wind direction is within 45° .

For each birth, I calculate the start of pregnancy as

$$C_i = DOB_i - 7 \times G_i$$

where DOB_i is the date of birth and G_i is the LMP-based gestational length in weeks. I

refer to C_i as the conception date throughout the analysis. Then, I assign every race in the sample period to a trimester relative to that pregnancy such that Trimester 1 (T1) covers conception through gestational day 97; T2 covers days 98-195; and T3 covers days 196-280, following ACOG conventions (ACO, 2024). Each mother is matched to the nearest of the four Texas tracks, and the analysis sample is restricted to mothers whose nearest track is TMS and who reside within 25 km of it. I construct exposure at the birth-by-race-day level, recording for each race whether it fell within the pregnancy, in which trimester, how much lead it emitted, and whether the mother was downwind that day, and then aggregate these to one observation per birth. The unit of analysis is therefore the birth.

Table 2. Summary Statistics: Births Within 25 km of TMS

	Pre-transition		Post-transition		Full sample	
	mean	sd	mean	sd	mean	sd
Birthweight (g)	3358.7530	(539.2535)	3339.0232	(534.8890)	3348.3073	(537.0081)
Low birthweight (<2500g)	0.0469	(0.2114)	0.0513	(0.2206)	0.0495	(0.2168)
Gestational age (weeks)	38.4821	(1.7845)	38.4709	(1.7923)	38.4769	(1.7872)
Apgar score (5 min)	8.9083	(0.5491)	8.8794	(0.6524)	8.8939	(0.6082)
Gestational hypertension	0.0431	(0.2031)	0.0431	(0.2031)	0.0437	(0.2043)
Eclampsia	0.0014	(0.0372)	0.0006	(0.0236)	0.0009	(0.0299)
Gestational diabetes	0.0220	(0.1467)	0.0357	(0.1856)	0.0298	(0.1699)
Maternal age (years)	28.7188	(5.8572)	28.5265	(5.8846)	28.6033	(5.8664)
Married	0.8048	(0.3964)	0.7522	(0.4318)	0.7785	(0.4152)
Education	0.4069	(0.4913)	0.3854	(0.4867)	0.3940	(0.4886)
WIC enrollment	0.2438	(0.4294)	0.2675	(0.4427)	0.2577	(0.4373)
Black	0.0453	(0.2080)	0.0537	(0.2254)	0.0497	(0.2173)
White	0.8761	(0.3295)	0.8539	(0.3532)	0.8613	(0.3456)
First birth	0.4051	(0.4909)	0.4096	(0.4918)	0.4083	(0.4915)
Insured	0.8858	(0.3180)	0.8904	(0.3125)	0.8858	(0.3180)
Distance to TMS (km)	19.1596	(4.6090)	18.7322	(4.9143)	18.9310	(4.7711)
Races during T1	1.1535	(1.0320)	1.6225	(1.2654)	1.4043	(1.1770)
Races during T2	1.2183	(1.0627)	1.5863	(1.2688)	1.4216	(1.1838)
Races during T3	1.0699	(1.0267)	1.4133	(1.2669)	1.2525	(1.1847)
Mean fetal blood lead, T1	0.2672	(0.3304)	0.0000	(0.0000)	0.1621	(0.2855)
Mean fetal blood lead, T2	0.4296	(0.3261)	0.0000	(0.0000)	0.2396	(0.3174)
Mean fetal blood lead, T3	0.4944	(0.3626)	0.0000	(0.0000)	0.2499	(0.3441)
AUC, full pregnancy	110.3115	(55.7193)	0.0000	(0.0000)	60.6137	(66.8430)
Peak blood lead	2.4432	(0.9465)	0.0000	(0.0000)	1.3388	(1.3550)
Observations	32437		28620		72471	

Notes: Mean with standard deviation in parentheses. Pre-transition is defined as conception before December 17, 2005; post-transition as conception on or after March 1, 2007. Full sample includes all births in the study area (including the transition window excluded from the pre/post columns). Sample consists of all Texas births from 2003–2010 whose nearest racetrack is TMS and who reside within 25 km. Education is defined as having at least a college degree.

4 Exposure Construction

I construct several measures of race-related lead exposure during pregnancy. The primary measure is based on the amount of lead emitted during races that occur in each trimester of pregnancy. Distance from the racetrack and race-day wind direction are secondary measures used to capture spatial variation in exposure. Finally, I use a pregnancy-specific physiologically based pharmacokinetic (PBPK) model to account for the dynamics of lead within the maternal body.

4.1 Emissions-Based Exposure

For each birth-race pair, I compute the lead emitted during the race and then sum it over the races that fall in each trimester and over the full pregnancy. Lead emitted at race j is calculated as:

$$\text{LeadEmitted}_j = \frac{\text{MilesTraveled}_j}{\text{MPG}} \times \text{LeadPerGallon} \quad (1)$$

where LeadEmitted_j is the total amount of lead in grams released during race j ; MilesTraveled_j is the total miles driven by all cars during race j ; MPG is miles per gallon and is assumed to equal five which is a typical fuel economy for NASCAR; and LeadPerGallon is set to 5.2 grams, the average pre-transition tetraethyllead concentration per gallon ([Hollingsworth et al., 2025](#)). Before the transition, Equation 1 measures the lead actually released during a race. After the transition, races used unleaded fuel and released no lead, so I apply the same formula to construct a counterfactual measure of the lead each race would have emitted had it still used leaded fuel.

For each mother i and trimester $k \in \{1, 2, 3\}$, I sum the total amount of lead emitted from Equation 1 across all races at TMS whose date falls within trimester k of her pregnancy:

$$E_{ik} = \sum_{j \in \text{trimester}_k(i)} \text{LeadEmitted}_j. \quad (2)$$

Because the distribution is right-skewed, I take the log plus one transformation,

$$\tilde{E}_{ik} = \log(1 + E_{ik}). \quad (3)$$

where $\tilde{E}_{ik}$ is the log emission for mother i in trimester k . Under the hypothesis that the pre-transition effect operates through lead, the post-transition coefficient on $\tilde{E}_{ik}$ should be

near zero. However, a meaningful post-transition effect would suggest that non-lead factors, such as noise, traffic, or residential sorting near the track, may be driving the result. If so, attributing the pre-transition effect specifically to lead would be unwarranted.

4.2 Distance and Wind Direction

Distance and wind direction provide two sources of variation in exposure in this study. First, a mother who lives closer to the racetrack should experience greater exposure to lead than a mother who lives farther away. Therefore, the effect of lead exposure on birth-weight should attenuate with distance from the racetrack. For each mother i , I compute distance, d_i , in kilometers from her census tract centroid to TMS and restrict the sample to mothers living within 25 km of TMS. Second, I classify mother i as living “downwind” of TMS on each race day if the angle between the bearing from TMS to her census tract centroid and the observed wind direction at the nearest weather station is within $\pm 45^\circ$.

For each trimester k of mother i ’s pregnancy, I construct two downwind measures. The first measure is the fraction of trimester k race days on which she was downwind of TMS,

$$\text{ShareDownwind}_{ik} = \frac{\#\{\text{trimester } k \text{ race days with } i \text{ downwind}\}}{\#\{\text{trimester } k \text{ race days}\}}, \quad (4)$$

defined as zero for mothers with no race days in trimester k and one for mothers who reside downwind on every race day in trimester k . If mother i was downwind of the race-track on two of three race days during trimester k , for instance, then her downwind exposure share would be 67 percent. The second is a binary measure that captures whether a mother was downwind of TMS on at least one race day during the trimester:

$$\text{AnyDownwind}_{ik} = 1 [i \text{ is downwind on at least one trimester } k \text{ race day}], \quad (5)$$

If she resides downwind on at least one race day in trimester k , then AnyDownwind_{ik} equals one. It is possible for mother i to reside downwind on all three of her trimesters.

Downwind classification on a race day

Mother A's bearing from TMS falls within the $\pm 45^\circ$ plume cone (downwind);
Mother B's bearing falls outside (upwind).

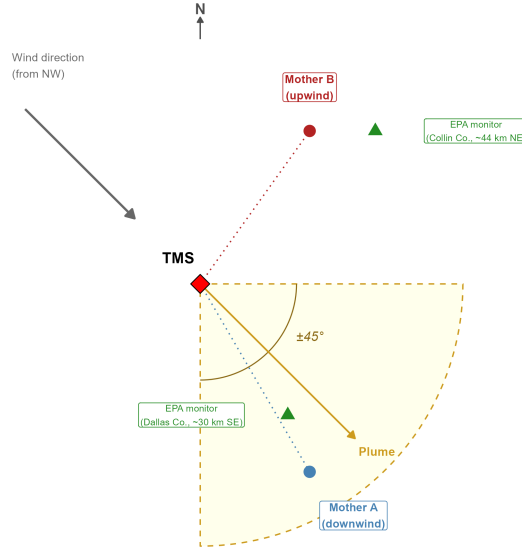


Figure 3. On each race day, mother i is classified as residing “downwind” of TMS if the angle between (a) the bearing from TMS to her census-tract centroid and (b) the observed wind direction at the nearest weather station that day falls within $\pm 45^\circ$.

4.3 Pregnancy-Specific PBPK Model

A central challenge in this setting is that lead exposure occurs intermittently. Lead has a half-life of approximately 30 to 40 days in blood, several weeks in soft tissue, and years to decades in bone. Thus, exposure occurring late in one trimester can elevate maternal and fetal blood lead well into the next, even if a mother is not exposed to a race during the subsequent trimester. Moreover, lead stored in maternal bone may be slowly remobilized into the blood during the later stages of pregnancy (Télez-Rojo et al., 2004). Hence, the reduced-form specification (Equation 3) cannot capture these intra-pregnancy dynamics. It identifies whether a mother was exposed and the trimester in which the exposure occurred, but it does not capture how lead from that exposure persists in the maternal body across subsequent trimesters.

To account for this, I develop a pregnancy-specific physiologically based pharmacokinetic (PBPK) model to capture the accumulation and redistribution of lead across gestation. The model follows the general PBPK framework in O’Flaherty (1991); Oflaherty (1993, 1995) but simplifies its full physiological structure and underlying lead kinetics into three maternal compartments: blood, soft tissue, and bone. When mother i is exposed to lead released during a race, the dose enters her blood as a discrete pulse on the gestational day on which the race occurs. I define this dose using the race’s lead emissions from

Equation 1, expressed relative to the average leaded race occurring during pregnancy in the sample:

$$q_{ij} = \frac{\text{LeadEmitted}_j \times \text{Leaded}_j}{\overline{\text{LeadEmitted}}}, \quad \overline{\text{LeadEmitted}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \text{LeadEmitted}_j \quad (6)$$

where LeadEmitted_j is the amount of lead, in grams, released during race j from Equation 1; Leaded_j equals one if race j used leaded fuel and zero otherwise; and $\mathcal{P}$ is the set of all mother-race pairs in which a leaded race occurs during the mother's pregnancy. $\overline{\text{LeadEmitted}}$ is therefore the average emissions of a leaded race in the full sample, weighted by the number of pregnancies each race overlaps. This makes q_{ij} a normalized dose expressed in units of an average leaded race. A value of $q_{ij} = 2$, for example, means that race j emitted twice as much lead as the average leaded race while races using unleaded fuel have a dose of zero. Since the dose is based only on race-level emissions, all mothers exposed to the same race receive the same dose in the PBPK model. Race-day exposure is then represented as

$$E_i(t) = \sum_{j=1}^{J_i} q_{ij} \delta(t - t_{ij}) \quad (7)$$

where $E_i(t)$ is the total lead input from race-day exposure on gestational day t ; J_i is the number of races during mother i 's pregnancy; q_{ij} is the normalized lead dose for race j ; t_{ij} is the gestational day on which race j occurs; and $\delta(\cdot)$ is the Dirac delta function. Thus, each race contributes a single exposure pulse on the day it occurs.

Let $B(t)$, $S(t)$, and $K(t)$ denote lead in blood, soft tissue, and bone, respectively.³ The dynamics of lead across the three compartments are given by

$$\dot{B}(t) = E(t) + k_{KB}(t)K(t) + k_{SB}S(t) - (k_{BS} + k_{BK} + k_{B0})B(t) \quad (8)$$

$$\dot{S}(t) = k_{BS}B(t) - k_{SB}S(t) \quad (9)$$

$$\dot{K}(t) = k_{BK}B(t) - k_{KB}(t)K(t) \quad (10)$$

where $E(t)$ is the race-day lead input entering maternal blood on gestational day t , equal

³For ease of notation, I suppress the subscript i .

to the dose q_{ij} on the day of a race and zero otherwise; k_{B0} is the rate at which blood lead is eliminated through excretion; k_{BS} and k_{SB} are the rates of transfer between blood and soft tissue; and k_{BK} and $k_{KB}(t)$ are the rates of transfer between blood and bone.

Lead stored in maternal bone can be released into the blood during bone resorption. Maternal bone turnover changes over gestation, and skeletal lead can contribute to maternal blood lead, particularly later in pregnancy (Téllez-Rojo et al., 2004; Manton et al., 2003). Because the model simplifies the O’Flaherty framework into a single bone compartment, I approximate the bone-to-blood transfer rate, $k_{KB}(t)$, to increase over gestation. I parameterize $k_{KB}(t)$ as

$$k_{KB}(t) = \begin{cases} k_{KB}^{(1)} = 0.0003 & 0 \leq t \leq 97 \\ k_{KB}^{(2)} = 0.00045 & 98 \leq t \leq 195 \\ k_{KB}^{(3)} = 0.0006 & 196 \leq t \leq 280 \end{cases} \quad (11)$$

The values of $k_{KB}(t)$ are calibrated parameters to capture the increase in skeletal lead mobilization over pregnancy. I set the baseline rate to $k_{KB} = 0.0003$ per day, approximately the daily equivalent of a 10 percent annual skeletal turnover rate (Manolagas and Parfitt, 2010). I allow this rate to increase across trimesters to reflect changes in maternal bone resorption during pregnancy. In particular, Curtis et al. (2021) examine C-terminal telopeptide of type I collagen (CTX), a biochemical marker of bone resorption that is released during the breakdown of type I collagen in bone. They find that median maternal urinary CTX approximately doubled between 14 and 34 weeks of gestation. Therefore, I scale the baseline bone-to-blood transfer rate by 1, 1.5, and 2 across the first, second, and third trimesters, respectively, yielding $k_{KB} = 0.00030, 0.00045$, and 0.00060 per day. The remaining transfer rates are also calibrated using the human stable-isotope tracer study of Rabinowitz et al. (1976), which estimates lead movement across three compartments corresponding approximately to blood, soft tissue, and bone. The EPA translates their estimates into daily transfer rates of approximately 0.0079 from blood to soft tissue, 0.0033 from soft tissue back to blood, 0.0037 from blood to bone, and 0.019 from blood to excretion (U.S. Environmental Protection Agency, 2001). Hence, I set the parameters as $k_{BS} = 0.0079$, $k_{SB} = 0.0033$, $k_{BK} = 0.0037$, and $k_{B0} = 0.019$ per day. Solving the system using the Euler method with a one-day time step, I set the initial conditions $B(0) = S(0) = K(0) = 0$ and simulate lead levels from gestational day 0 to day 280.⁴ The system produces a daily maternal blood-lead trajectory, $B_i(t)$, which I use as a proxy for

⁴The state variables capture lead attributable to race-day exposure rather than the mother’s total lead burden. The zero initial conditions therefore indicate no race-attributable lead at the start of pregnancy.

fetal lead exposure given that lead readily crosses the placenta and lead levels in maternal and umbilical cord blood are highly correlated (Zhu et al., 2010).

To construct trimester-specific exposure measures, I average $B_i(t)$ over the gestational days in each trimester. Note that $B_i(t)$ is expressed in normalized exposure units rather than micrograms per deciliter. Let T_k denote the set of gestational days in trimester $k \in \{1, 2, 3\}$, where $T_1 = \{0, \dots, 97\}$, $T_2 = \{98, \dots, 195\}$, and $T_3 = \{196, \dots, 280\}$, and let $|T_k|$ denote the number of days in T_k . Mean predicted fetal blood lead for mother i in trimester k is

$$\bar{B}_{ik} = \frac{1}{|T_k|} \sum_{t \in T_k} B_i(t) \quad (12)$$

To measure cumulative exposure over pregnancy, I sum $B_i(t)$ across gestational days to obtain the area under the maternal blood-lead trajectory (AUC):

$$\text{AUC}_i = \sum_{t=1}^{280} B_i(t). \quad (13)$$

I also measure peak exposure as the maximum daily maternal blood-lead level over pregnancy:

$$\text{Peak}_i = \max_{t \in \{1, \dots, 280\}} B_i(t) \quad (14)$$

Hence, the trimester means $\bar{B}_{i1}, \bar{B}_{i2}, \bar{B}_{i3}$, the cumulative-dose measure AUC_i , and the peak Peak_i describe the simulated fetal blood lead trajectory based on when the exposure occurred, how much lead was accumulated overall, and how high it spiked.

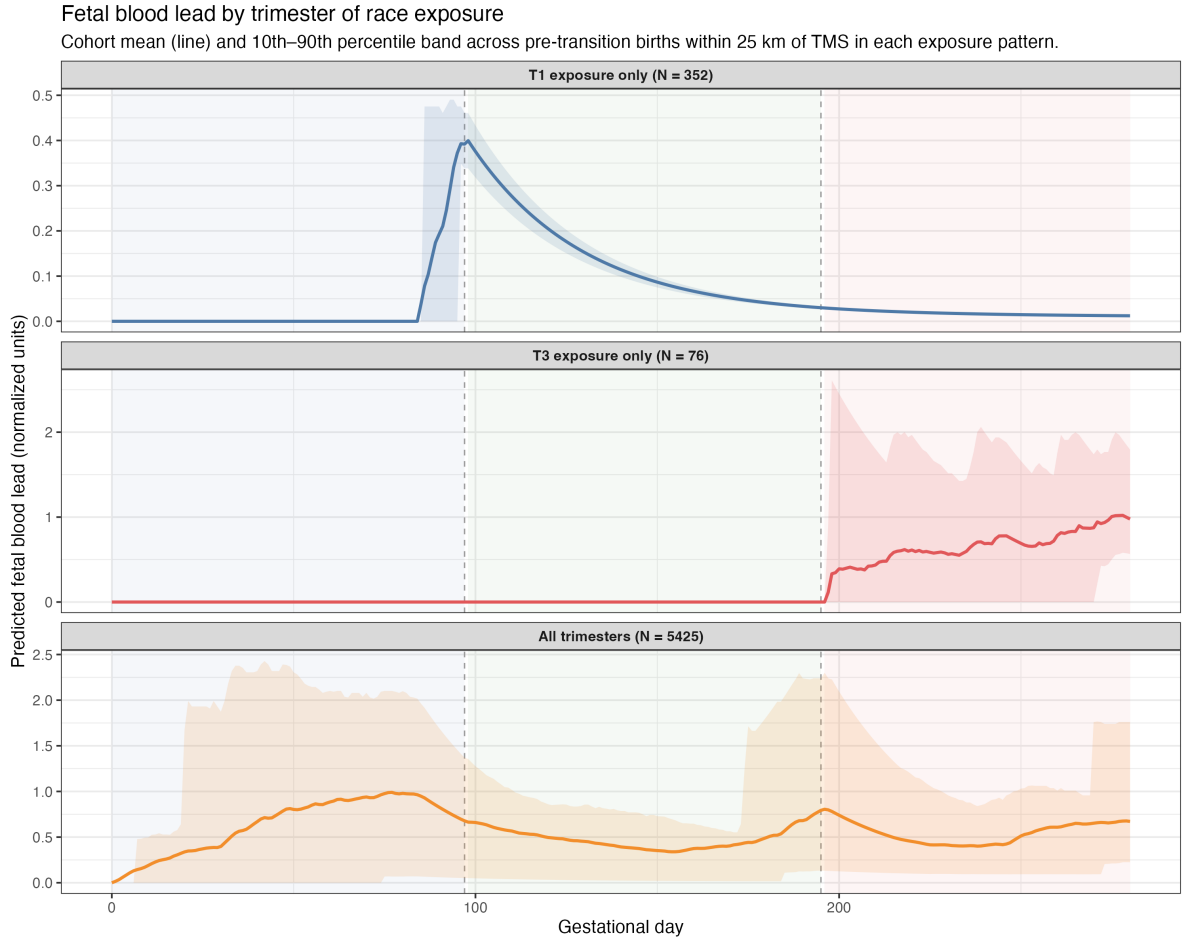


Figure 4. The top panel shows mothers living within 25 km of TMS whose only leaded race exposures occurred in the first trimester ($N = 352$); the middle panel shows mothers exposed only in the third trimester ($N = 76$); the bottom panel shows mothers exposed in all three trimesters ($N = 5,425$).

5 Empirical Strategy

5.1 Within-TMS Identification

The identification strategy relies on comparing births conceived before and after the 2006/2007 switch to unleaded gasoline at TMS while exploiting variation in the timing, location, and intensity of race-day exposure. The sample is restricted to births within 25 km of TMS occurring between 2003 and 2010. Because race dates are fixed, two mothers who conceive just a few weeks apart can be exposed to the same races, but those races can fall in different gestational windows. Additionally, because lead concentrations decay with distance from the track, mothers who live closer to the racetrack are likely to experience greater exposure than those living farther away. The spatial gradient test (Section 6.3) examines the relationship between birthweight and distance from the track. Fur-

thermore, two mothers can have very different exposures depending on whether their residences fall within the wind plume on race days. Finally, the level of emissions released during a race varies with factors such as accidents and race length. This variation is plausibly unrelated to maternal health. Appendix A tests whether log emissions in each trimester are correlated with predetermined maternal characteristics: maternal age, marital status, WIC enrollment, race, first birth, and insurance status. Of the 21 coefficients, only one is statistically significant at the 5 percent level (second-trimester emissions and WIC enrollment), and the remaining coefficients are small and statistically insignificant.

Since all mothers live within 25 km of TMS and experience at least one race during pregnancy, exposure varies across mothers depending on how many races fall within each trimester. Hence, there is no control group in the traditional sense. Mothers with zero races in a given trimester provide the within-period baseline. Exploiting variation in conception timing relative to the fixed calendar of TMS races, I estimate the effect of an additional race occurring in each trimester separately for the pre- and post-transition periods. If the pre-transition relationship is driven by lead exposure, the estimated effects should be negative before the transition and attenuate toward zero after the transition. This pattern would support the interpretation that the pre-transition effects are driven by lead rather than by other factors associated with racing.

5.2 Baseline Specification

The baseline reduced-form specification is estimated separately for the pre- and post-transition samples:

$$BW_{ict} = \alpha + \sum_{k=1}^3 \beta_k Z_{ik} + X'_{ict} \gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \quad (15)$$

where BW_{ict} is birthweight in grams for birth i in census tract c at calendar time t ; Z_{ik} is the log trimester-specific exposure measure ($\tilde{E}_{ik}$ in Equation 3); X_{ict} is a vector of maternal demographic, socioeconomic, and health controls (marital status, WIC enrollment, chronic hypertension, education, race, first birth, insurance status, and maternal age group)⁵; μ_c are census tract fixed effects; μ_y are birth-year fixed effects; and μ_m are birth-month fixed

⁵WIC enrollment and insurance status are not recorded on the birth certificate before 2005. For births in 2003-04, I set both to zero and include an indicator for missing values, which is absorbed by the birth-year fixed effects. The coefficients on WIC and insurance are therefore identified from births in 2005 onward, and births in 2003-04 remain in the sample. I control for chronic hypertension rather than a broader index of maternal risk factors because it is recorded consistently throughout the birth certificates compared to other conditions like diabetes.

effects. Standard errors are clustered at the census tract level.

I estimate Equation (15) separately for the pre- and post-transition samples using the same log emissions measure, $\tilde{E}_{ik}$. Before the transition, it measures the lead actually emitted during races. After the transition, it measures the amount of lead that would have been emitted if the race had used the same leaded fuel as before the 2006/2007 switch. The pre-transition β_k captures the relationship between lead emissions during trimester k and birthweight, whereas the post-transition β_k captures the relationship between the corresponding counterfactual emissions measure and birthweight after races no longer emitted lead.

Similarly, the specification for wind direction is:

$$BW_{ict} = \alpha + \sum_{k=1}^3 \theta_k \text{ShareDownwind}_{ik} + \sum_{k=1}^3 \rho_k R_{ik} + X'_{ict} \gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \quad (16)$$

where BW_{ict} is birthweight in grams for birth i in census tract c at calendar time t ; $\text{ShareDownwind}_{ik}$ is the fraction of trimester k race days on which mother i was downwind of TMS (Equation 4); R_{ik} is the number of races at TMS during trimester k of mother i 's pregnancy; θ_k the coefficient on the downwind share in trimester k ; ρ_k controls for the overall number of races in that trimester; X_{ict} is the same vector of controls as in Equation (15); μ_c are census tract fixed effects; μ_y are birth-year fixed effects; and μ_m are birth-month fixed effects. Holding the number of races during the trimester fixed, θ_k measures the difference in birthweight associated with an increase in the share of race days on which the mother was downwind of TMS. A negative θ_k indicates that mothers who were downwind on a larger share of race days experienced greater plume exposure and had infants with lower birthweight.

6 Results

This section presents the paper's main findings. During the pre-transition period, prenatal exposure to race activity at Texas Motor Speedway is associated with lower birthweight, shorter gestation, and a higher probability of low birthweight, with the largest effects concentrated among births from 2005 onward. These relationships are substantially weaker or absent after the transition to unleaded fuel. Consistent with spatial decay in airborne lead exposure, the reduction in birthweight is larger among mothers living closer to the racetrack before the transition, while this distance gradient disappears afterward. Mothers living downwind of the racetrack on race days also deliver infants with

lower birthweight, although the post-transition wind estimates are not uniformly consistent with the lead hypothesis. The PBPK model further shows that higher simulated fetal blood-lead exposure, particularly during the second trimester, is associated with lower birthweight. Finally, the results provide suggestive evidence that gestational hypertension may be one pathway through which prenatal lead exposure affects birth outcomes.

6.1 Pre-Transition vs. Post-Transition Effect

Table 3 reports the estimated effects of prenatal exposure to race activity at TMS on birthweight (grams), low birthweight (<2,500 grams), gestational age (weeks), and birthweight conditional on gestational age (grams). Each specification is estimated separately for the pre- and post-transition periods. Because racing continued after the transition without leaded fuel, the post-transition period provides a placebo test for race-related exposures that persisted after the transition.

Panel A reports the estimates using the logarithm of lead emissions. Before the transition, a doubling of emissions during pregnancy is associated with a 0.024-week reduction in gestational age, approximately 0.17 days, ($-0.0353 \times \ln(2)$) and a 0.26-percentage-point increase in the probability of low birthweight ($0.0037 \times \ln(2)$). Both estimates are statistically significant while the corresponding birthweight estimate is small and statistically insignificant. The trimester-specific birthweight estimates are statistically insignificant in both periods.

Panel B reports the estimates based on the number of races during pregnancy. Before the transition, each additional race is associated with lower birthweight across all three trimesters, with the largest reduction occurring in the second trimester. One additional race in the second trimester reduces birthweight by 31.1 grams, or 0.9 percent of the average birthweight in the pre-transition period (3,359 grams). Over the full pregnancy, each additional race is associated with a 13.8-gram reduction in birthweight. After the transition, the estimates are substantially smaller and statistically insignificant across all trimesters.

The reductions in birthweight are accompanied by shorter gestation. Before the transition, one additional race in the second trimester is associated with a 0.15-week reduction in gestational age, or approximately one day, while the post-transition estimates are close to zero across all trimesters. Similarly, each additional race in the second trimester raises the probability of low birthweight by 0.99 percentage points, which is a 21 percent increase relative to the pre-transition prevalence of 4.7 percent. However, the estimate remains positive and statistically significant post-transition in the second trimester,

which provides less clear evidence of a lead-specific effect on low birthweight. The estimates for birthweight conditional on gestational age also show no consistent pre/post pattern. Additionally, race exposure in the first trimester is associated with lower conditional birthweight in both periods while the full-pregnancy estimates are statistically insignificant. The next subsection examines whether these relationships differ across cohorts with greater variation in race exposure.

Table 3. Main Reduced-Form Results

Panel A: Log emissions (T1+T2+T3 joint)								
	Pre-transition (leaded)				Post-transition (as-if-leaded, placebo)			
	(1) BW (g)	(2) LBW	(3) Gest. age (wks)	(4) BW GA	(1) BW (g)	(2) LBW	(3) Gest. age (wks)	(4) BW GA
Log emissions, T1	1.6552 (1.2453)	-0.0009* (0.0005)	0.0165*** (0.0046)	-0.8865 (0.9949)	-1.9450 (1.2825)	0.0007 (0.0007)	0.0036 (0.0056)	-2.5933*** (0.7928)
Log emissions, T2	-0.1231 (1.3141)	-0.0001 (0.0006)	0.0009 (0.0056)	-0.2682 (1.2096)	-0.7607 (1.4618)	0.0009 (0.0006)	0.0029 (0.0058)	-1.2703 (0.9648)
Log emissions, T3	-1.2878 (1.4530)	-0.0001 (0.0006)	-0.0087 (0.0055)	0.0591 (1.1374)	0.5727 (1.3201)	-0.0001 (0.0006)	0.0077* (0.0043)	-0.8021 (0.9141)
Observations	32437	32437	32437	32437	28620	28620	28620	28620
R-squared	0.0289	0.0132	0.0169	0.2846	0.0323	0.0118	0.0184	0.3845
<i>Log emissions, full pregnancy</i>								
Log emissions, full pregnancy	-3.8014 (4.7563)	0.0037* (0.0019)	-0.0353** (0.0167)	1.6341 (3.7548)	1.3226 (19.5117)	0.0001 (0.0095)	0.0916 (0.0816)	-15.0467 (11.8246)
Observations	32437	32437	32437	32437	28620	28620	28620	28620
R-squared	0.0288	0.0131	0.0162	0.2845	0.0322	0.0116	0.0183	0.3843
Panel B: Cumulative race count (T1+T2+T3 joint)								
	BW (g)	LBW	Gest. age (wks)	BW GA	BW (g)	LBW	Gest. age (wks)	BW GA
Races during T1	-18.9421*** (6.0610)	0.0034 (0.0022)	-0.0538** (0.0257)	-10.6542* (5.8771)	-7.5301 (6.4298)	0.0036 (0.0032)	0.0109 (0.0282)	-9.4820*** (3.3909)
Races during T2	-31.0589*** (6.6549)	0.0099*** (0.0027)	-0.1469*** (0.0325)	-8.4336 (6.1272)	-6.7829 (6.8896)	0.0075** (0.0030)	-0.0171 (0.0286)	-3.7205 (4.7966)
Races during T3	-14.8449** (5.7044)	0.0035* (0.0021)	-0.0903*** (0.0231)	-0.9363 (4.7320)	1.4051 (5.5334)	0.0019 (0.0028)	0.0062 (0.0195)	0.3039 (4.4441)
Observations	32437	32437	32437	32437	28620	28620	28620	28620
<i>Cumulative race count, full pregnancy</i>								
Total Races, Full Pregnancy	-13.7853*** (4.7813)	0.0021 (0.0017)	-0.0588*** (0.0186)	-4.7267 (4.2701)	-1.7435 (4.5410)	0.0013 (0.0024)	0.0162 (0.0202)	-4.6407 (2.9607)
Observations	32437	32437	32437	32437	28620	28620	28620	28620
R-squared	0.0292	0.0130	0.0167	0.2846	0.0322	0.0117	0.0183	0.3843
Standard errors in parentheses. Sample: TMS live births within 25 km. All regressions include census tract and year and month fixed effects plus standard maternal controls. Standard errors clustered by tract. Outcomes: birthweight in grams (BW), low birthweight <2500g (LBW), gestational age in weeks, and birthweight conditional on gestational age (BW GA). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.								

6.2 Heterogeneity by Birth Cohort

In the pre-transition period, TMS hosted one major race weekend each year, typically in the spring, along with two smaller truck races. Beginning in 2005, TMS added a second Cup and Xfinity race weekend in the fall. This matters for identification because the

addition of a second race weekend creates variation in exposure across years that is not absorbed by birth-month fixed effects.

Table 4 re-estimates the pre-transition specifications separately for births in 2003-04 and for births from 2005 onward. Among births from 2005 onward, the estimated effects are larger in magnitude than those in the pooled pre-transition sample. One additional race during the second trimester reduces birthweight by 43.5 grams (1.3 percent of the mean), raises the probability of low birthweight by 1.42 percentage points, and shortens gestation by 0.22 weeks. The emissions-based estimates also show a similar pattern. A doubling of emissions during pregnancy reduces birthweight by 42.6 grams, raises the probability of low birthweight by 1.41 percentage points (a 29 percent increase relative to the 4.8 percent prevalence in these cohorts), and shortens gestation by 0.19 weeks (or about 1.3 days).

In contrast, the estimates for births in 2003 and 2004 are smaller and generally statistically insignificant. The race-count estimates have standard errors two to three times larger than those for the later cohorts. Although the pre-transition results are primarily driven by the 2005–2006 cohorts, two findings support interpreting these estimates as effects of lead exposure rather than the schedule change itself. First, the birthweight and gestational-age estimates are close to zero after the transition, even though TMS maintained the same two-weekend schedule. Second, the effects in the later cohorts are present for both exposure measures: race counts and log emissions. However, I cannot distinguish the effects of greater lead exposure under the two-weekend schedule from other changes specific to cohorts born around 2006.

Table 4. Pre-Transition Results by Birth Year

Panel A: Log emissions (T1+T2+T3 joint)								
	Births 2003–04 (spring weekend only)				Births 2005 on (spring + fall weekends)			
	(1) BW (g)	(2) LBW	(3) Gest. age (wks)	(4) BW GA	(1) BW (g)	(2) LBW	(3) Gest. age (wks)	(4) BW GA
Log emissions, T1	1.5980 (2.2002)	-0.0001 (0.0009)	0.0137 (0.0083)	-0.2935 (2.0629)	1.5985 (1.3098)	-0.0016** (0.0007)	0.0209*** (0.0056)	-2.0045** (0.9694)
Log emissions, T2	2.8918 (1.8831)	0.0001 (0.0008)	0.0122 (0.0080)	1.2048 (1.8930)	-3.3194* (1.9560)	0.0001 (0.0010)	-0.0125 (0.0077)	-1.1705 (1.5141)
Log emissions, T3	0.9386 (2.2736)	-0.0007 (0.0009)	-0.0008 (0.0090)	1.0471 (1.6224)	-3.9029** (1.7340)	0.0004 (0.0008)	-0.0173** (0.0067)	-0.9210 (1.4260)
Observations	16725	16725	16725	16725	15712	15712	15712	15712
R-squared	0.0321	0.0162	0.0169	0.2327	0.0317	0.0140	0.0250	0.3557
<i>Log emissions, full pregnancy</i>								
Log emissions, full pregnancy	8.0910 (4.8913)	0.0005 (0.0019)	0.0166 (0.0165)	5.7983 (4.3313)	-61.4876*** (10.2737)	0.0204*** (0.0052)	-0.2754*** (0.0488)	-14.1008* (7.3215)
Observations	16725	16725	16725	16725	15712	15712	15712	15712
R-squared	0.0321	0.0161	0.0164	0.2327	0.0329	0.0148	0.0259	0.3556
Panel B: Cumulative race count (T1+T2+T3 joint)								
	BW (g)	LBW	Gest. age (wks)	BW GA	BW (g)	LBW	Gest. age (wks)	BW GA
Races during T1	-9.6466 (15.5466)	-0.0007 (0.0051)	0.0177 (0.0678)	-12.0868 (16.6888)	-21.7898*** (5.9124)	0.0047 (0.0030)	-0.0738*** (0.0270)	-9.0924** (4.2876)
Races during T2	1.4650 (17.2398)	0.0008 (0.0060)	-0.0129 (0.0747)	3.2419 (15.7772)	-43.5379*** (7.5761)	0.0142*** (0.0038)	-0.2161*** (0.0324)	-6.3435 (6.5336)
Races during T3	8.1478 (15.4526)	-0.0032 (0.0054)	-0.0327 (0.0586)	12.6513 (12.6804)	-22.8099*** (7.0307)	0.0056* (0.0031)	-0.1244*** (0.0309)	-1.3921 (5.5405)
Observations	16725	16725	16725	16725	15712	15712	15712	15712
R-squared	0.0320	0.0162	0.0165	0.2329	0.0329	0.0148	0.0270	0.3557
<i>Cumulative race count, full pregnancy</i>								
Total Races, Full Pregnancy	-1.0196 (12.9387)	-0.0028 (0.0044)	-0.0070 (0.0395)	-0.0604 (11.7342)	-18.5536*** (4.9118)	0.0035 (0.0026)	-0.0786*** (0.0264)	-5.0306 (3.6992)
Observations	16725	16725	16725	16725	15712	15712	15712	15712
R-squared	0.0319	0.0161	0.0164	0.2326	0.0319	0.0137	0.0238	0.3556

Standard errors in parentheses. Sample: pre-transition TMS live births within 25 km (conceived before 17 December 2005), split by birth year. Births from 2005 on include a small number born in early 2007. TMS added a second Cup and Xfinity race weekend in November 2005; before then it held one large spring weekend a year plus two truck races. All regressions include census tract and year and month fixed effects plus standard maternal controls. Standard errors clustered by tract. Outcomes: birthweight in grams (BW), low birthweight <2500g (LBW), gestational age in weeks, and birthweight conditional on gestational age (BW | GA).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3 Spatial Gradient Test

If airborne lead is driving the relationship, birthweight should increase with distance from the track in the pre-transition period as lead concentrations decay with distance, while the birthweight gradient should be approximately flat after the transition. Figure 5 shows that predicted birthweight increases with distance from TMS before the transition. This is consistent with the fact that mothers who reside closer to TMS are exposed to higher concentrations of lead and deliver infants with lower birthweight. Once lead is removed from the fuel, the distance gradient becomes approximately flat. This pattern is consistent with lead exposure driving the pre-transition relationship rather than other

race-related factors that persisted after the transition.

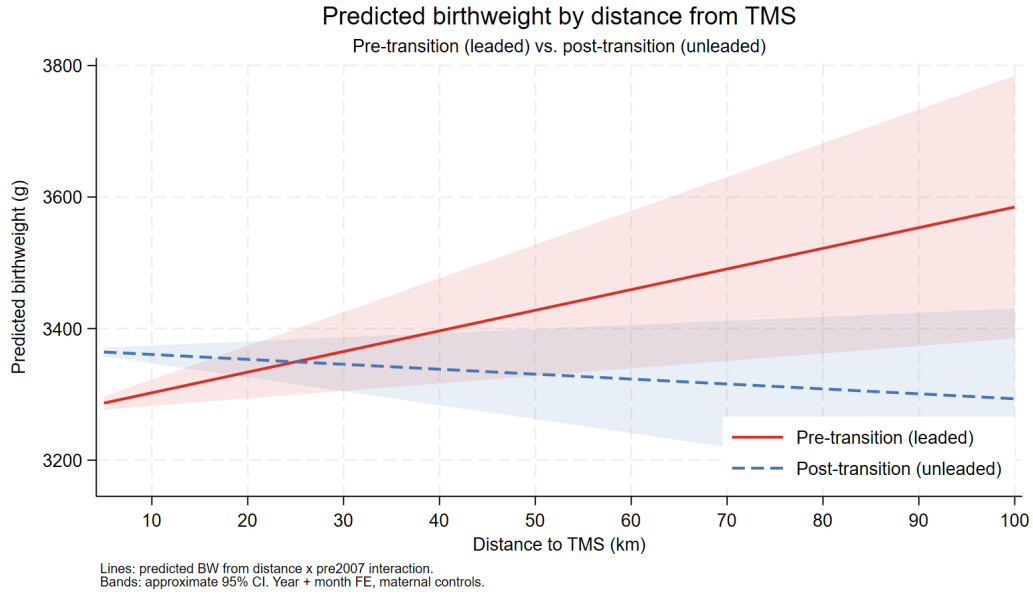


Figure 5. Predicted birthweight by distance from TMS. (Pre-transition = solid red; Post-transition = dashed blue). Birthweight is regressed on distance $\times$ pre-transition with year and month fixed effects and maternal controls. Shaded regions are approximate 95% confidence intervals.

Before the transition, birthweight increases by 1.25 grams for each additional kilometer from the track (significant at the 5 percent level). After the transition, the distance gradient is -0.66 grams per kilometer and statistically insignificant. The difference between the two slopes, given by the interaction between the pre-transition indicator and distance, is 1.91 grams per kilometer and is significant at the 5 percent level. Across the interquartile range of distance in the pooled sample (15.5 to 22.5 kilometers), the pre-transition gradient implies that infants born to mothers living farther from the track weighed about 9 grams more than those born to mothers living closer to it.

6.4 Wind Direction

Wind direction is plausibly exogenous to maternal sorting around the racetrack. Hence, mothers who reside downwind of the track on race days should experience larger birthweight reductions than those upwind, holding fixed the number of races during the trimester. Table 5 reports the relationship between downwind exposure and birthweight. Panel A uses the share of trimester k race days on which the mother was downwind, while Panel B uses a binary indicator for whether the mother was downwind on at least one race day during the trimester. All specifications control for the number of races in

each trimester, so the downwind coefficients capture the additional association between being in the path of emissions and birthweight.

In Panel A, mothers who were downwind on every race day in their first trimester delivered infants weighing 31.5 grams less, on average, than mothers who were never downwind. Similarly, Panel B shows that being downwind on at least one race day is associated with a 26.7-gram reduction in birthweight, or 0.8 percent of mean birthweight, relative to mothers exposed to the same number of races who were never downwind. In the third trimester, being downwind on at least one race day is associated with a 21.4-gram reduction in birthweight.

After the transition, the first- and second-trimester downwind estimates are close to zero in both panels, consistent with the removal of lead from racing fuel. However, the third-trimester estimates are positive and statistically significant, with Panel B showing a 22.6-gram increase in birthweight. This unexpected positive association complicates the interpretation of the third-trimester results.

Table 5. Wind Direction: Downwind Exposure and Birthweight

<i>Panel A: Share of race days downwind</i>		
	Pre-transition (1)	Post-transition (2)
Share downwind, T1	-31.5491** (13.1541)	13.4726 (17.4167)
Share downwind, T2	-8.8788 (11.0935)	11.3902 (20.9503)
Share downwind, T3	-17.9220 (11.3395)	22.9682* (13.6508)
Races during T1	-17.1806*** (5.9346)	-8.5070 (6.3822)
Races during T2	-31.2906*** (6.6166)	-7.5538 (7.3902)
Races during T3	-14.7895*** (5.3436)	-0.0206 (5.4602)
Observations	32437	28620
R-squared	0.0298	0.0323
<i>Panel B: Any downwind indicator</i>		
	Pre-transition	Post-transition
Any downwind, T1	-26.7388** (10.3401)	2.3288 (9.8403)
Any downwind, T2	-9.2338 (8.5972)	0.5182 (11.5096)
Any downwind, T3	-21.4144** (9.6409)	22.6362*** (7.4176)
Races during T1	-15.6948** (5.9804)	-7.8236 (6.3354)
Races during T2	-30.4740*** (6.6356)	-6.6355 (7.5525)
Races during T3	-13.2599** (5.5759)	-1.7006 (5.5628)
Observations	32437	28620
R-squared	0.0298	0.0325

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km. All regressions include census tract and year and month fixed effects, standard maternal controls, and cumulative race counts (shown). Standard errors clustered by tract. The share downwind measure is the fraction of a trimester's race days on which the mother's residence was downwind ($\pm 45^\circ$). The Any Downwind indicator equals one if the mother was downwind on at least one race day. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5 PBPK Dose-Response

The reduced-form estimates in Sections 6.1 to 6.4 treat each race as an equal and instantaneous shock. Lead, however, accumulates in the mother's body. The PBPK model captures this accumulation by simulating, for each mother, the daily path of blood lead over her full pregnancy. Because each race's dose is normalized by the average lead emitted per leaded race, one unit of simulated blood lead corresponds to the increase in maternal blood lead. The coefficients are therefore effects of relative rather than absolute blood-lead levels, and I interpret them by comparing mothers at different points of the simulated distribution. Since blood lead from racing is mechanically zero after the transition, the model reports estimates from the pre-transition period only.

Table 6 reports the results. Second-trimester blood lead has the strongest association with birthweight. Entered on its own, a one-unit increase in mean fetal blood lead during the second trimester lowers birthweight by 89.2 grams. Holding blood lead in the other trimesters constant (Column 4), the estimate is 132.2 grams. To put this in context, moving a mother from the 25th to the 75th percentile of second-trimester blood lead (0.16 to 0.72) lowers birthweight by about 73 grams, or 2.2 percent of the pre-transition mean. In the third trimester, blood lead also lowers birthweight (−59.6 grams per unit), which translates to a reduction of about 35 grams across its interquartile range. The first-trimester estimate is smaller (−36.5 grams).

Similarly, each additional unit of cumulative blood lead over the pregnancy lowers birthweight by 0.57 grams, so a mother at the 75th percentile of cumulative exposure delivers an infant about 33 grams lighter than a mother at the 25th percentile. Peak blood lead is also negatively associated with birthweight (−23.4 grams per unit). However, peak exposure varies relatively little across mothers.

Table 6. PBPK Dose-Response: Birthweight and Modeled Fetal Blood Lead (Pre-Transition)

	(1) T1 only	(2) T2 only	(3) T3 only	(4) Joint	(5) AUC	(6) Peak
Mean fetal blood lead, T1	-11.1734 (16.9177)			-36.5189** (17.6888)		
Mean fetal blood lead, T2		-89.1876*** (19.6344)		-132.1869*** (23.8559)		
Mean fetal blood lead, T3			-27.5071* (15.7111)	-59.5717*** (18.1239)		
AUC, full pregnancy					-0.5690*** (0.1407)	
Peak blood lead						-23.3704*** (6.2186)
Observations	32323	32323	32323	32323	32323	32323
R-squared	0.0288	0.0298	0.0290	0.0306	0.0298	0.0295

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km, pre-transition only. All regressions include census tract and year and month fixed effects, plus standard maternal controls. Standard errors clustered by tract. (Table 3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.6 Mechanism: Maternal-Vascular Pathway

The results in Table 3 indicate that most of the birthweight reduction operates through shorter gestation rather than slower fetal growth. Lead is an established cause of elevated blood pressure in adults as it induces oxidative stress, reduces nitric oxide, and increases vascular resistance (Navas-Acien et al., 2007). During pregnancy, elevated blood lead has been linked to gestational hypertension and preeclampsia (Poropat et al., 2018), which can reduce blood flow to the placenta. If lead from racing fuel raised maternal blood pressure, it would therefore shorten gestation and lower birthweight, which is the pattern observed in the main results. Furthermore, gestational hypertension is defined by onset after 20 weeks of pregnancy, so any effect of lead exposure should appear in the late second and third trimesters rather than the first.

Table 7 estimates the main specifications with gestational hypertension, eclampsia, the five-minute Apgar score, and an indicator for a low Apgar score (below 7). These outcomes are recorded consistently only on the revised birth certificate that Texas adopted in 2005. The five-minute Apgar score is not recorded before 2005, and gestational hypertension rises from 3.1-3.9 percent in 2003-04 to 5.1-5.4 percent in 2005-06, which reflects more how those conditions were reported rather than the incidence frequency. I therefore estimate the pre-transition mechanism regressions on births from 2005 onward.

Before the transition, third-trimester exposure raises the probability of gestational hypertension. Each additional race during the third trimester increases the probability by 0.90 percentage points (significant at the 1 percent level), which is a 17 percent increase relative to the 5.2 percent prevalence in these cohorts. Similarly, the third-trimester log emissions raise the probability of hypertension by 0.12 percentage point (significant at the 5 percent level). The concentration in the third trimester matches the clinical onset of gestational hypertension as well as the reductions in gestational age in the third trimester reported in Table 3.

The evidence for this channel is not uniform across specifications. First-trimester log emissions are negatively associated with hypertension, and exposure over the full pregnancy has no statistically significant effect. The post-transition estimates also are negative and marginally significant in the second and third trimesters (-0.49 and -0.32 percentage points). A negative association between race activity and hypertension when no lead was present is not predicted by the lead hypothesis, and I do not interpret it as evidence for it. The hypertension results are therefore suggestive. They point to a biologically plausible channel that is consistent with the results for gestational age.

The remaining outcomes provide little additional evidence. Eclampsia is rare, affecting fewer than 1 in 1,000 births in these cohorts. The effect is positive and significant in the first trimester, although eclampsia typically develops later in pregnancy. Moreover, each additional race in the first trimester is also associated with a 0.018-point reduction in the five-minute Apgar score on its 10-point scale and a 0.26-percentage-point increase in the probability of a low Apgar score, significant at the 5 and 10 percent levels, respectively. The effects disappear after the transition. These estimates are small, and the evidence for effects on the newborn's condition at birth is less clear than that for gestational age.

Table 7. Mechanism: Maternal-Vascular and Newborn Outcomes

<i>Panel A: Log emissions (T1+T2+T3 joint)</i>								
	Pre-transition (leaded), births 2005 on				Post-transition (as-if-leaded, placebo)			
	(1) Hyp.	(2) Ecl.	(3) Apgar5	(4) Low Apgar	(1) Hyp.	(2) Ecl.	(3) Apgar5	(4) Low Apgar
Log emissions, T1	-0.0018*** (0.0006)	0.0001 (0.0001)	-0.0027* (0.0016)	0.0005* (0.0003)	0.0002 (0.0005)	-0.0000 (0.0001)	-0.0003 (0.0015)	0.0000 (0.0003)
Log emissions, T2	-0.0003 (0.0009)	-0.0000 (0.0001)	-0.0036** (0.0017)	0.0004 (0.0003)	-0.0007 (0.0005)	-0.0000 (0.0001)	-0.0009 (0.0016)	-0.0001 (0.0003)
Log emissions, T3	0.0018** (0.0007)	0.0001 (0.0002)	0.0025 (0.0023)	-0.0002 (0.0003)	-0.0005 (0.0004)	-0.0000 (0.0001)	0.0009 (0.0015)	-0.0003 (0.0003)
Observations	15712	15712	15132	15132	28620	28620	27904	27904
<i>Log emissions, full pregnancy</i>								
Log emissions, full pregnancy	0.0072 (0.0049)	0.0016 (0.0009)	0.0038 (0.0140)	-0.0009 (0.0025)	-0.0068 (0.0064)	0.0001 (0.0009)	0.0246 (0.0308)	-0.0036 (0.0059)
Observations	15712	15712	15132	15132	28620	28620	27904	27904
R-squared	0.0129	0.0080	0.0100	0.0064	0.0122	0.0041	0.0115	0.0058
<i>Panel B: Cumulative race count (T1+T2+T3 joint)</i>								
	Hyp.	Ecl.	Apgar5	Low Apgar	Hyp.	Ecl.	Apgar5	Low Apgar
Races during T1	-0.0027 (0.0024)	0.0011** (0.0004)	-0.0175** (0.0085)	0.0026* (0.0015)	-0.0006 (0.0025)	-0.0001 (0.0003)	0.0019 (0.0094)	-0.0003 (0.0015)
Races during T2	0.0043 (0.0032)	0.0009* (0.0005)	-0.0148 (0.0090)	0.0012 (0.0016)	-0.0049* (0.0025)	-0.0002 (0.0003)	0.0012 (0.0094)	-0.0008 (0.0014)
Races during T3	0.0090*** (0.0029)	0.0008 (0.0006)	0.0159* (0.0083)	-0.0020 (0.0014)	-0.0032* (0.0016)	0.0000 (0.0003)	0.0058 (0.0078)	-0.0010 (0.0014)
Observations	15712	15712	15132	15132	28620	28620	27904	27904
<i>Cumulative race count, full pregnancy</i>								
Total Races, Full Pregnancy	0.0030 (0.0022)	0.0009* (0.0005)	0.0018 (0.0069)	0.0001 (0.0012)	-0.0010 (0.0016)	0.0000 (0.0002)	0.0048 (0.0079)	-0.0006 (0.0015)
Observations	15712	15712	15132	15132	28620	28620	27904	27904
R-squared	0.0129	0.0083	0.0100	0.0064	0.0122	0.0041	0.0115	0.0058

Standard errors in parentheses. Outcomes (LPM for binary): gestational hypertension (Hyp.), eclampsia (Ecl.), Apgar 5-min score, low Apgar indicator (<7). Sample: TMS live births within 25 km. Pre-transition columns use births from 2005 on, recorded on the revised birth certificate: the Apgar score is not recorded before 2005, and recorded gestational hypertension and eclampsia change with the 2005 certificate revision. All regressions include census tract and year and month fixed effects plus standard maternal controls. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Conclusion

This paper examines the effects of prenatal lead exposure on birth outcomes by exploiting NASCAR's and ARCA's transition from leaded to unleaded racing fuel in 2006–2007. I find that prenatal lead exposure is associated with lower birthweight, shorter gestation, and a higher probability of low birthweight. These effects are most pronounced during the second trimester and are substantially weaker or absent after the transition to unleaded fuel. The results also provide suggestive evidence that gestational hypertension may be one pathway through which lead exposure affects birth outcomes. Before the

transition, mothers living closer to the racetrack and those residing downwind on race days deliver infants with lower birthweight. The distance gradient also disappears after the transition while the wind results provide additional, though less uniform, support for a lead-specific effect. Because racing continued after lead was removed from the fuel, the pre- and post-transition comparison helps distinguish lead exposure from other factors, such as traffic and noise. The pregnancy-specific PBPK model further supports the lead effect. Its estimates show that moving from the 25th to the 75th percentile of simulated blood lead in the second trimester is associated with an approximately 73-gram reduction in birthweight. The EPA monitor-based construction in the Appendix also finds negative effects between lead exposure and birthweight. These findings suggest that localized and intermittent sources of airborne lead can adversely affect maternal and infant health. As lead remains present today in sources such as piston-engine aircraft and older housing paints, the results suggest that further reductions in exposure may produce health benefits for nearby communities, particularly for pregnant women and their infants.

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Appendix

A Balance Test

Table 8 reports the balance test referenced in Section 5.1. Each column is a separate regression of predetermined maternal characteristics on joint log emissions with census tract and year and month fixed effects on the pre-transition sample.

Table 8. Balance Test: Log Emissions vs. Maternal Characteristics (Pre-Transition)

	(1) Mat. age	(2) Married	(3) WIC	(4) Black	(5) White	(6) First birth	(7) Insured
Log emissions, T1	0.0022 (0.0108)	0.0001 (0.0006)	0.0016 (0.0011)	-0.0002 (0.0004)	0.0007 (0.0006)	0.0013 (0.0009)	-0.0002 (0.0008)
Log emissions, T2	0.0071 (0.0115)	0.0004 (0.0009)	0.0028** (0.0014)	0.0003 (0.0004)	0.0007 (0.0006)	0.0010 (0.0007)	0.0003 (0.0010)
Log emissions, T3	-0.0028 (0.0107)	0.0008 (0.0006)	0.0014 (0.0009)	0.0001 (0.0004)	0.0004 (0.0007)	-0.0008 (0.0008)	-0.0003 (0.0009)
Observations	32437	32437	15712	32437	32437	32437	15712
R-squared	0.1342	0.0889	0.1646	0.0251	0.0330	0.0104	0.1066

Standard errors in parentheses. Each column is a separate regression of the indicated predetermined maternal characteristic on joint log emissions (T1+T2+T3) during pregnancy. Sample: TMS live births within 25 km, pre-transition (conception before Dec 17, 2005). Fixed effects: census tract, birth year, and birth month. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Monitor-Based Exposure

For each race day, I compute an inverse distance weighted (IDW) ambient lead concentration at each mother's residence using all EPA monitors within 150 km:

$$w_{im} = \frac{1}{\max(d_{im}, 1)}, \quad \text{Lead}_{ij} = \frac{\sum_m w_{im} \cdot \text{ExcessLead}_{mj}}{\sum_m w_{im}} \quad (17)$$

where w_{im} is the inverse-distance weight assigned to monitor m for mother i ; d_{im} is the distance in kilometers between mother i 's census tract centroid and monitor m (monitors beyond 150 km receive zero weight); ExcessLead_{mj} is the race-day excess lead concentration at monitor m on race day j (the monitor's reading minus its annual non-race-day baseline); and Lead_{ij} is the IDW ambient lead concentration assigned to mother i on race day j measured in $\mu\text{g}/\text{m}^3$.

The race-day absorbed dose for mother i on race day j is then:

$$q_{ij}^A = a \cdot \text{Lead}_{ij} \cdot IR_i \cdot \Delta t_{ij} \quad (18)$$

where q_{ij}^A is the absorbed lead dose (measured in micrograms) for mother i on race day j ; Lead_{ij} is the IDW ambient lead concentration from Equation 17; $a = 0.3$ is the standard pulmonary absorption fraction (Oflaherty, 1993); $IR_i = 16 \text{ m}^3/\text{day}$ is the maternal inhalation rate (U.S. Environmental Protection Agency, 2011); and $\Delta t = 1$ day represents the 24-hour averaging period of the monitor readings. For mothers without monitor coverage during a given trimester, the assigned exposure is set to zero.

Table 9. EPA AQS lead monitors at TMS: race-day coverage 2003–2010

Site ID	County	Distance	Years active	Total readings	Race days (pre)	Race days (post)	Mean excess Pb	Consistent
480850009	Collin	43.7	2003–2010	469	3	3	0.000	✓
480850003	Collin	44.0	2003–2010	463	3	3	0.000	✓
480850007	Collin	44.0	2003–2010	463	3	3	0.000	✓
481130057	Dallas	47.8	2003–2007	1393	14	0	0.023	✓
481130018	Dallas	56.9	2003–2007	239	3	0	0.024	✓
481130061	Dallas	47.1	2003–2003	8	0	0	–	
481130066	Dallas	57.2	2003–2004	101	1	0	0.046	

Notes: These EPA Air Quality System (AQS) ambient lead monitors are within ≈ 150 km of Texas Motor Speedway with at least one valid reading during 2003–2010. Distance is the distance from the monitor’s reported coordinates to TMS. “Years active” is the year in which the monitor reported any reading, regardless of whether the reading fell on a race day. “Total readings” is the count of all valid daily readings the monitor reported over 2003–2010.

“Race days (pre)” counts the number of TMS race days on which this monitor reported a reading restricted to race days through Dec. 17, 2005; “Race days (post)” covers race days from Mar. 1, 2007 onward. “Mean excess Pb” is the monitor’s mean race-day reading minus its annual non-race-day baseline, averaged over its pre-transition race day (in $\mu\text{g}/\text{m}^3$); “–” indicates the monitor recorded no pre-transition race-day readings.

“Consistent” comprises monitors that operated continuously throughout the pre-transition window (2003–2006) and is used in the robustness check reported in Section B.2. Site 481130066 is excluded because it dropped out after 2004. Site 481130061 is excluded because it operated only briefly in 2003 and caught no race days. Of the 17 pre-transition race-day readings in the consistent set, 14 come from site 481130057 in Dallas County.

B.1 Monitor Calibration

Table 10 reports the results in three panels. Panel A reports the absorbed dose by trimester. In the joint specification, the first and second trimesters have the largest effects. Similarly, Panel B reports the PBPK-simulated trimester-mean blood lead alongside the full-pregnancy cumulative dose (AUC), and the first and second trimesters dominate which is different from Table 6, where the second and third trimesters dominate. Panel C reports the cumulative race count by trimester and the full-pregnancy total. While the results differ by trimester, the monitor-based exposure and emission-based exposure suggest that higher prenatal lead exposure reduces birthweight.

Table 10. Model A Calibration: Monitor-Based Exposure (Pre-Transition)

<i>Panel A: Absorbed dose (monitor-based)</i>				
	(1) T1 only	(2) T2 only	(3) T3 only	(4) Joint
Absorbed dose, T1	-57.2945*** (15.8970)			-79.0309*** (18.4230)
Absorbed dose, T2		-31.2113** (13.6701)		-63.2642*** (16.0127)
Absorbed dose, T3			-25.4409 (17.0327)	-21.4470 (17.2043)
Observations	32437	32437	32437	32437
R-squared	0.0292	0.0289	0.0288	0.0297
<i>Panel B: PBPK trimester means and cumulative dose</i>				
	T1 only	T2 only	T3 only	Joint
Mean fetal blood lead, T1	-222.0128*** (59.8711)			-251.7994*** (63.2801)
Mean fetal blood lead, T2		-254.5222*** (64.1690)		-281.2876*** (66.2393)
Mean fetal blood lead, T3			-126.2436** (53.7870)	-54.8924 (54.8655)
Observations	32437	32437	32437	32437
Cumulative dose (AUC)	-1.8748*** (0.3739)			
Observations	32437			
R-squared	0.0297			
<i>Panel C: Cumulative race count</i>				
	T1 only	T2 only	T3 only	Joint
Races during T1	-2.3064 (4.5542)			-18.9421*** (6.0610)
Races during T2		-10.0689** (4.5628)		-31.0589*** (6.6549)
Races during T3			-2.9784 (5.1252)	-14.8449** (5.7044)
Observations	32437	32437	32437	32437
Races, full pregnancy	-13.7853*** (4.7813)			
Observations	32437			
R-squared	0.0292			

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km, pre-transition only. Fixed effects: census tract, year, month. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Consistent-Monitor Robustness

A potential concern with using EPA monitors is that EPA monitor coverage near TMS changed over the sample period. Of the seven monitors used in the monitor-based specification, two dropped out during the pre-transition window (site 481130066 after 2004 and site 481130061 after 2003). This could make the inference fragile. To address monitor attrition, I re-estimate the model's regressions using only the five monitors active throughout the entire pre-transition window (2003–2006) and compare the results side by side. Table 11 reports the birthweight in grams.

In Panel A, the full-monitor coverage produces a coefficient of -79.0 while the consistent-monitor coverage produces a coefficient of -74.4 in the first trimester, all significant at the one percent level. The same pattern holds for the other trimesters. In Panel B, the full-monitor coverage produces a coefficient of -281.3 while the consistent-monitor coefficient is -271.4 in the second trimester, again both significant at the one percent level. In Panel C, the full-monitor coefficient is -1.9 and the consistent-monitor coefficient is -1.8 , both significant at the one percent level. The standard errors are nearly identical across specifications. This stability rules out monitor attrition as a driver of the model's result.

Table 11. Consistent-Monitor Robustness

<i>Panel A: Absorbed dose (joint, pre-transition)</i>		
	(1) Full (7 monitors)	(2) Consistent (5 monitors)
Absorbed dose, T1	-79.0309*** (18.4230)	-74.3965*** (17.3625)
Absorbed dose, T2	-63.2642*** (16.0127)	-61.7322*** (15.7701)
Absorbed dose, T3	-21.4470 (17.2043)	-21.1151 (17.3216)
Observations	32437	32437
R-squared	0.0297	0.0297
<i>Panel B: PBPK trimester means (joint, pre-transition)</i>		
	Full (7 monitors)	Consistent (5 monitors)
Mean fetal blood lead, T1	-251.7994*** (63.2801)	-227.2672*** (59.1541)
Mean fetal blood lead, T2	-281.2876*** (66.2393)	-271.4331*** (64.8403)
Mean fetal blood lead, T3	-54.8924 (54.8655)	-55.6525 (55.1844)
Observations	32437	32437
R-squared	0.0299	0.0299
<i>Panel C: Cumulative dose (pre-transition)</i>		
	Full (7 monitors)	Consistent (5 monitors)
Cumulative dose (AUC)	-1.8748*** (0.3739)	-1.8133*** (0.3597)
Observations	32437	32437
R-squared	0.0297	0.0297

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km, pre-transition only. Fixed effects: census tract, year, month. Standard errors clustered by tract. Full (7 monitors) uses 7 EPA monitors within 100 km of TMS with race-day coverage during 2003–2010. Consistent (5 monitors) restricts to the 5 monitors active throughout 2003–2006. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Threats to Cross-Track Comparisons

NASCAR and ARCA also held races at venues other than TMS. Thunder Hill Raceway hosted five regional races between 2005 and 2009, while Meyer Speedway and Texas World Speedway are legacy racetracks with no NASCAR activity during the sample period. Table 12 examines whether using these venues in a cross-track comparison would

affect identification relative to the within-TMS design. Panel A reports the results for Thunder Hill Raceway. Each additional race during the second trimester is associated with birthweight that is 57 grams higher before the transition and 58 grams higher after it. The post-transition estimates are positive and significant in every trimester. Thunder Hill hosted only five regional races in which lead status cannot be determined, making it an unreliable comparison site for identifying the effects of lead exposure. In Panel B, Meyer Speedway exhibits a negative and statistically significant post-transition coefficient of approximately 19 grams, whereas Texas World Speedway shows no significant change. The decline in birthweight at Meyer Speedway may reflect a shock in the Houston area after 2007 rather than NASCAR-related factors. It could potentially be driven by local demographic and economic changes around the 2008 financial crisis. Given that there is no similar decline at Texas World Speedway, the change may be specific to the Houston area rather than reflecting a general effect of lead exposure. These results highlight a potential problem with using births near Meyer as a control group for those near TMS. The within-TMS design avoids this issue by restricting the sample to mothers living near TMS and exploiting variation in race exposure within census tracts and across birth years and months. This approach addresses threats to cross-track comparisons that may be affected by different regional trends.

Table 12. Replication and Falsification at Other Texas Tracks

<i>Panel A: Thunder Hill — log emissions (pre — post)</i>		
	(1) Pre-transition	(2) Post-transition (as-if-leaded, placebo)
Log emissions, T1	5.5167 (3.6335)	10.3370*** (1.5649)
Log emissions, T2	8.1016*** (1.9257)	8.2880** (3.1387)
Log emissions, T3	-2.6762 (4.8623)	7.2911* (3.9121)
Observations	8192	9023
R-squared	0.0455	0.0318
Log emissions, full pregnancy	2.7490 (2.7561)	8.8824*** (1.8470)
Observations	8192	9023
R-squared	0.0451	0.0318
<i>Panel A (cont.): Thunder Hill — cumulative race count (pre + post)</i>		
	Pre-transition	Post-transition (placebo)
Races during T1	39.0137 (25.6955)	72.5631*** (11.0268)
Races during T2	57.2939*** (13.6184)	57.9096** (22.5210)
Races during T3	-18.9258 (34.3859)	51.5090* (27.4380)
Observations	8192	9023
R-squared	0.0455	0.0318
Total Races, Full Pregnancy	19.4410 (19.4909)	62.2996*** (13.4863)
Observations	8192	9023
R-squared	0.0451	0.0317
<i>Panel B: Post-2007 birthweight break at legacy tracks (falsification)</i>		
	Meyer (Houston)	TX World (College Stn.)
Post-transition	-19.4498*** (4.7123)	5.2111 (16.2220)
Year trend	2.9271** (1.3402)	-3.2126 (5.1923)
Observations	291563	16295
R-squared	0.0400	0.0469

Standard errors in parentheses. Outcome: birthweight in grams. City fixed effects due to the small geographic area. Sample: births within 25 km. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Race Schedule

Table 13 lists every NASCAR and ARCA race at the four Texas tracks during the sample period (2003-2010). TMS hosted multiple national-series races each year, typically clustering around two weekends in April and November. Thunder Hill hosted a small number of KNPSW regional races between 2005 and 2009. Texas World Speedway and Meyer Speedway had no races during the sample period and are omitted.

Table 13. NASCAR Race Schedule at Texas Tracks, 2003–2010

Track	Date	Series	Fuel	Miles	Drivers
Thunder Hill	10/01/2005	KNPSW	Leaded	1132	20
Thunder Hill	09/30/2006	KNPSW	Leaded	1156	22
Thunder Hill	03/31/2007	KNPSW	Unleaded	1472	28
Thunder Hill	04/19/2008	KNPSW	Unleaded	1188	22
Thunder Hill	03/14/2009	KNPSW	Unleaded	974	19
TMS	03/29/2003	Xfinity	Leaded	10236	43
TMS	03/30/2003	Cup	Leaded	19646	43
TMS	06/06/2003	CWTS	Leaded	6538	36
TMS	10/11/2003	CWTS	Leaded	6537	36
TMS	04/03/2004	Xfinity	Leaded	10576	43
TMS	04/04/2004	Cup	Leaded	19016	43
TMS	06/11/2004	CWTS	Leaded	6651	34
TMS	10/16/2004	CWTS	Leaded	6568	36
TMS	04/16/2005	Xfinity	Leaded	11724	43
TMS	04/17/2005	Cup	Leaded	19006	43
TMS	06/10/2005	CWTS	Leaded	8067	36
TMS	11/04/2005	CWTS	Leaded	7262	36
TMS	11/05/2005	Xfinity	Leaded	11922	43
TMS	11/06/2005	Cup	Leaded	20862	43
TMS	04/08/2006	Xfinity	Leaded	12855	43
TMS	04/09/2006	Cup	Leaded	19456	43
TMS	06/09/2006	CWTS	Leaded	8409	36
TMS	11/03/2006	CWTS	Unleaded	7240	36
TMS	11/04/2006	Xfinity	Unleaded	11392	43
TMS	11/05/2006	Cup	Leaded	20432	43
TMS	04/14/2007	Xfinity	Unleaded	10738	43
TMS	04/15/2007	Cup	Unleaded	19338	43
TMS	06/08/2007	CWTS	Unleaded	6768	35
TMS	11/02/2007	CWTS	Unleaded	6540	36
TMS	11/03/2007	Xfinity	Unleaded	11133	43
TMS	11/04/2007	Cup	Unleaded	19131	43
TMS	04/05/2008	Xfinity	Unleaded	10794	43
TMS	04/06/2008	Cup	Unleaded	20853	43
TMS	06/06/2008	CWTS	Unleaded	7508	35
TMS	10/31/2008	CWTS	Unleaded	5216	34
TMS	11/01/2008	Xfinity	Unleaded	10635	43
TMS	11/02/2008	Cup	Unleaded	21098	43
TMS	04/04/2009	Xfinity	Unleaded	11266	43
TMS	04/05/2009	Cup	Unleaded	19972	43
TMS	06/05/2009	CWTS	Unleaded	5848	33
TMS	11/06/2009	CWTS	Unleaded	5968	36
TMS	11/07/2009	Xfinity	Unleaded	8836	43
TMS	11/08/2009	Cup	Unleaded	18784	43
TMS	04/16/2010	Arca	Unleaded	4350	38
TMS	04/19/2010	Cup	Unleaded	19040	43
TMS	04/19/2010	Xfinity	Unleaded	10138	43
TMS	06/04/2010	CWTS	Unleaded	5962	36
TMS	11/05/2010	CWTS	Unleaded	6591	36
TMS	11/06/2010	Xfinity	Unleaded	10760	43
TMS	11/07/2010	Cup	Unleaded	18435	43

All NASCAR and ARCA races at the four Texas tracks during the sample period (2003–2010). TMS hosted Cup Series, Xfinity Series, and Camping World Truck Series races in most years, typically clustering around two weekends per year (April and November). Thunder Hill hosted five KNPSW regional races between 2005 and 2009. Texas World Speedway and Meyer Speedway hosted no races during the sample period (last active in 1993 and 1971, respectively) and are omitted from this table. Fuel status indicates whether the race used leaded or unleaded gasoline based on the series-specific transition dates described in Section 2.